



Dollar Movements and Dollar Dominance in the Aftermath of Liberation Day

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Abstract

This paper provides econometric evidence in support of the view that following President Trump's chaotic tariff announcements on Liberation Day, April 2, the dollar switched from being a safe-haven currency that appreciates in times of market volatility to a "risk-on" currency that moves inversely with volatility. We estimate an equation for daily changes in the DXY dollar index, using as explanatory variables daily changes in US-foreign interest rate differentials and the VIX, a measure of market volatility. We find a significant break in the relationship between the dollar and its primary determinants after Liberation Day, with the dollar falling below its predicted level. More importantly, in the two months after Liberation Day, the sensitivity of the dollar to the VIX shifted from positive to negative, suggesting that global investors ceased to treat the dollar as a safe haven in times of stress. Most recently, the dollar's sensitivity to the VIX has retraced some of its earlier decline, but whether this signals a return of the dollar's safe-haven status remains to be seen.

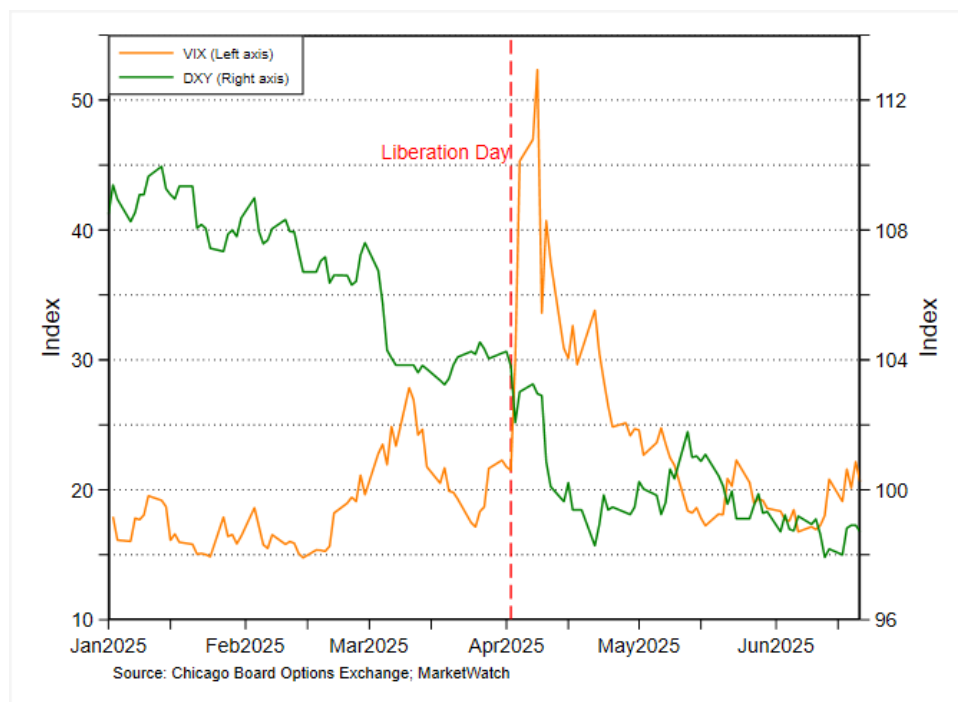
I. Introduction

President Trump's announcement of huge hikes in tariffs on April 2, which he labelled "Liberation Day," triggered severe financial turmoil. Especially jarring for financial market participants and onlookers was the fact, shown below, that the dollar declined sharply even as the VIX index, a measure of financial volatility, soared upward. In large part reflecting the safety and liquidity of U.S. Treasury securities, the dollar is a "flight-to-safety" currency, and generally rises during times of crisis. When the dollar fell instead of rising alongside the VIX after Liberation Day, this sparked widespread speculation that global investors, recoiling at the

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capricious and ill-considered policies of the Trump Administration, were abandoning the dollar and signalling the end of the currency's global dominance.

Fig 1. VIX Volatility Index and DXY Dollar Index, January 1–June 20, 2025



But does Liberation Day mark a turning point in the global stature of the dollar, or is the failure of the dollar to act as a flight-to-safety currency more of a temporary aberration, barring future assaults by the Administration on the stability of the global trade and financial system? To address this issue, we estimate an econometric model of daily dollar movements over the period January 2021 through June 20, 2025. We start with an equation for the *level* of the DXY dollar index, a measure of the foreign exchange value of the dollar against a weighted average of major advanced-economy currencies. We include as explanatory variables the VIX index and the interest rate differential between the US and foreign economies. We then estimate an equation for daily *changes* in the DXY dollar index, using as explanatory variables daily changes in interest rate differentials and the VIX, as well as the one-day-lagged level of residuals from

the levels regression described above. Thus, we essentially estimate a simple error-correction model for the dollar.

We find, first, that the estimation results conform closely to our theoretical expectations. Increases in US-foreign interest rate differentials lead to increases in the dollar, as does the VIX, confirming the behaviour of the dollar as a flight-to-safety currency. Second, on several days following Liberation Day, the dollar fell by a greater extent relative to the model's predictions than in the entire 2021-2025 sample period. This supports the widespread perception that a break in the relationship between the dollar and its primary determinants occurred after April 2. We provide statistical verification of a structural break in the model's parameters at that time.

Third and finally, we hone in on the estimated sensitivity of the dollar to the VIX: We find that this sensitivity, while generally positive over the course of the sample period (suggesting a long-standing flight-to-safety response), dropped to an historically outsized negative level in the two months after Liberation Day. This suggests that the safe-haven behaviour of the dollar evaporated, with the dollar switching from a "risk off" to a "risk on" currency. However, since the beginning of June, the sensitivity of the dollar has retraced some of its earlier decline, and as of the final observation for the model—Friday, June 20—it remains negative but insignificantly different from zero.

It is too soon to assess whether a lasting break in the dollar's flight-to-safety role has occurred. One possibility is that with markets continuing to recover from the shock of Liberation Day, additional observations will show the sensitivity of the dollar to the VIX climbing into positive territory, signalling the return of the dollar's safe-haven status. Another possibility is that the sensitivity of the dollar to the VIX will stabilize at its present value near zero,

indicating a loss of safe-haven status, but stopping short of becoming a permanent “risk-on” emerging market-style currency. Finally, it is possible that the response of the dollar depends on the source of the market’s volatility: Turbulence triggered by US policies may lower the dollar, while turbulence triggered by foreign geopolitical events—such as the eruption of the Iran-Israel-US hostilities observed in the past couple of weeks—may engender the dollar’s usual flight-to-quality behaviour. The weeks and months to come will shed more light on which of these possibilities is most likely.

A further consideration is that even if the safe-haven status of the dollar erodes, that does not signal the immediate demise of the dollar’s dominant role in the world financial system. The dollar is hardwired into the global economy. Inertia, network effects (people accept payments in dollars because everyone else accepts payments in dollars), and a vast financial and payments infrastructure ensure it will take many years for a challenger to dislodge the dollar’s pre-eminence, if one can even be found (Kamin and Sobel, 2024).

The plan of this paper is as follows. Section II places the paper in the context of prior research on this topic. Sections III and IV describe estimation results for the “levels” and “changes” versions of the model, respectively, while Section V describes the results’ robustness to a longer sample period. Section VI concludes.

II. Relevant Literature

There is an extensive body of research modelling the dollar exchange rate. We will not review that here – see Obstfeld and Zhou (2022) and Engel and Wu (2023) for details. Both of these papers estimate models relating changes in the level of the dollar to US-foreign interest rate differentials, measures of covered interest parity (CIP) deviations, measures of financial

market volatility, and the lagged real exchange rate.² We follow this approach below, although to streamline the model, we exclude CIP deviations (see footnote below) and replace the lagged real exchange rate with residuals from a levels model of nominal exchange rate.

The use of the sensitivity of the dollar to market volatility as a measure of the dollar's safe-haven status is also well-recognized in the literature. Fatum and Yamamoto (2016) note: "We define a safe haven currency as a currency that increases its relative value against other currencies as market uncertainty increases." This view is echoed by other analysts, including Rinaldo and Söderlind (2010), Habib and Stracca (2012), De Bock and de Carvalho Filho (2015), and Park and Fang (2025). These researchers generally find the US dollar to be a safe-haven currency, and sometimes also the Japanese yen and Swiss franc. They also frequently use the VIX as the measure of market uncertainty and volatility, and we follow that approach in our modelling.

As noted in the introduction, since Liberation Day, there has been considerable discussion in the financial media about how the apparent switch in the dollar from a "risk off" to "risk on" currency. Besides more anecdotal treatments of this issue, however, there have been only a couple of more analytical pieces. Hartley and Rebucci (2025) dispute that the fall in the dollar after Liberation Day owed to a weakening of its safe-haven status, arguing instead that worries about equity earnings prompted the retreat from dollar assets. Collyns and Klein (2025), however, estimate a simple model of the dollar as a function of US interest rates and a policy

² Deviations from covered interest parity are believed to reflect, in part, the special convenience or liquidity services offered by US assets. As described by Du, Tepper, and Verdelhan (2018), among others, these deviations reflect, among other things, changes in the demand for US Treasuries caused by changes in market volatility. We do not include them in the analysis in order to provide a cleaner and more complete read on the response of the dollar to market volatility as reflected in the VIX.

uncertainty variable; they find that policy uncertainty generally boosted the dollar except in two periods, January-May 2017 and December 2024-April 2025, consistent with a recent decline in the safe-haven status of the dollar. Our research follows along similar lines, but using a more fully developed model (eg, estimating in daily changes, using interest rate differentials) and focusing on market volatility, the most important dimension of safe-haven status.

III. A “levels” model of the dollar

Column (1) in the table below presents results of estimating a regression of the log-level of the DXY index of the dollar (multiplied by a hundred), a measure of its value against the currencies of other major advanced economies, on (1) the difference between US and a weighted average (using the same weights as the DXY index) of foreign 2-year Treasury yields, (2) the difference between the slope of the US yield curve (10-year minus 2-year) and the weighted average of foreign yield curve slopes, and (3) the VIX.³ The model fits well, and the coefficients on the interest rate differentials are significant and correctly signed.

³ The DXY dollar index is a weighted average of nominal dollar exchange rates against the euro, yen, Pound sterling, Canadian dollar, Swiss franc, and Swedish Krona. It is drawn from MarketWatch. US and foreign yields are drawn from the Board of Governors of the Federal Reserve System (US) via FRED and MarketWatch respectively. The VIX index is drawn from the Chicago Board Options Exchange via FRED®

Table 1: Regressions of the DXY Dollar on its Main Determinants

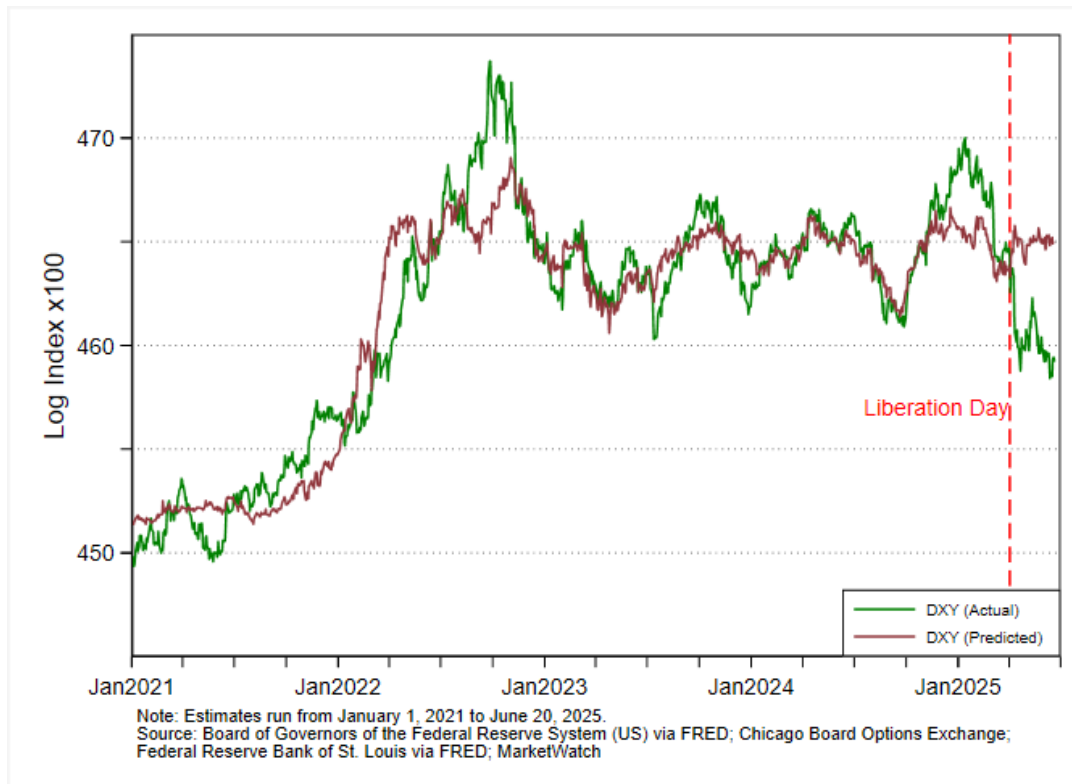
VARIABLES	(1)	(2)
	Log DXY Index x100	Log DXY Index x100 (daily changes)
	OLS	OLS
	2021-01-01 to 2025-06-20	2021-01-01 to 2025-06-20
2yr Yield Differential (level)	6.38*** (0.22)	
Yield Slope Differential (level)	0.81*** (0.29)	
CBOE Volatility Index: VIX (level)	0.02 (0.01)	
2yr Yield Differential (changes)		3.63*** (0.34)
Yield Slope Differential (changes)		1.49*** (0.46)
CBOE Volatility Index: VIX (changes)		0.04*** (0.01)
Reg (1) residual, 1-day lag (level))		-0.02*** (0.01)
Constant	452.49*** (0.47)	0.01 (0.01)
Observations	1,037	1,036
R-squared	0.87	0.18

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Figure 2 below compares the in-sample predictions (fitted values) of the model with the actual path of the log-DXY index. The model tracks the DXY reasonably well, though it does not capture all the turning points. Notably, the dollar shoots up well above the model in the months after the November election, likely reflecting the “Trump trade” in anticipation of his deregulatory pro-growth policies, and then it falls well below the model starting in February, well before Liberation Day. This suggests that even before Liberation Day, the bloom was coming off the rose regarding enthusiasm for Trump’s economic policies. The last downward jog in the actual dollar relative to predictions, even as the predicted value rebounded owing to higher Treasury yields and a surge in the VIX, was likely attributable to the investor dismay with Liberation Day and its aftermath, as noted in the introduction.

Figure 2: Actual and In-Sample Predicted Values of the DXY Dollar Based on Regression #1



More generally, however, movements in the foreign exchange value of the dollar should not be equated with changes in the dollar's status among global investors. The dollar has exhibited wide swings in its value over the past decades, even as it has maintained its dominance as an international reserve asset and vehicle currency. The fall in the dollar relative to model predictions in the months before Liberation Day may have been driven by any number of factors that are not well-captured by the model and, at the same time, not necessarily related to the role of the dollar in global trade and finance: a correction of excessive valuations

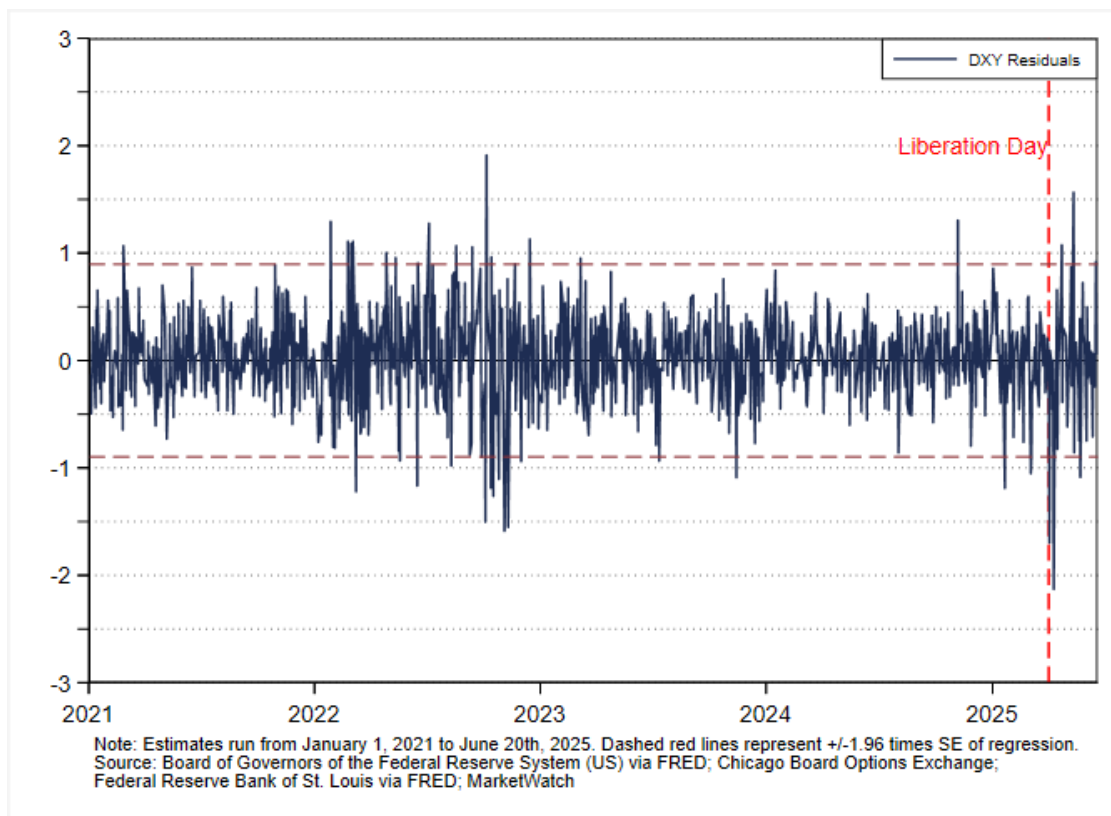
as the Trump trade lost appeal, or concerns that tariffs and other misguided policies would slow US economic growth.

IV. A model of daily dollar dynamics

The levels model shown in column (1) essentially captures the long-run relationship between the dollar and its determinants. It is ill-equipped to capture the dynamics of day-to-day movements in the dollar. To capture those dynamics, column (2) presents the estimations results of a regression of the daily *change* in the log of DXY on *changes* in the interest rate differentials, *changes* in the VIX, and, finally, the one-day-lagged level of the residuals from the levels model presented in column (1). The last term is an error-correction variable: when the actual dollar exceeds its prediction on the previous day (a positive residual), there is a tendency for it to decline back toward the model prediction, and when the dollar falls below its earlier prediction, there is a tendency for it to rise. So the coefficient on the lagged residual is expected to be negative.

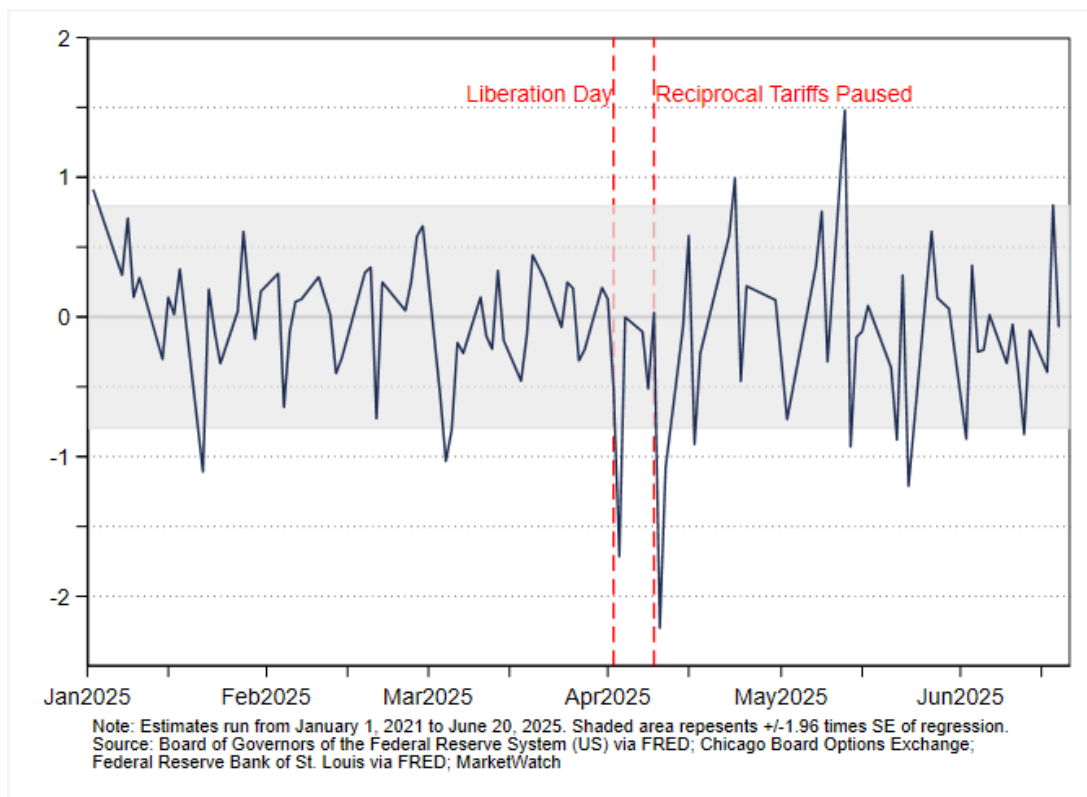
As shown in column (2), the coefficients on all the explanatory variables are correctly signed and statistically significant. Figure 3 presents the residuals from the regression #2 model, that is, the difference between the actual daily movements in the dollar and those predicted by the model, with the 95% confidence interval for these residuals outlined by the dashed horizontal lines. It shows that following “Liberation Day,” because the dollar declined while the VIX soared, changes in the dollar fell below their predicted values by the greatest margins in the entire sample period.

Fig 3. Daily Changes in the DXY Index of the Dollar: Residuals Based on Regression #2



The chart below hones in on just the past few months. The extreme residuals of the model are concentrated in just two days: April 3, after the tariffs were announced, and (ironically), April 10, following the retraction of most of the so-called “reciprocal tariffs”. However, even after that, there seem to have been more frequent large negative residuals than in the pre-Liberation Day months.

Fig 4. Daily Changes in the DXY Index of the Dollar in 2025: Residuals Based on Regression #2

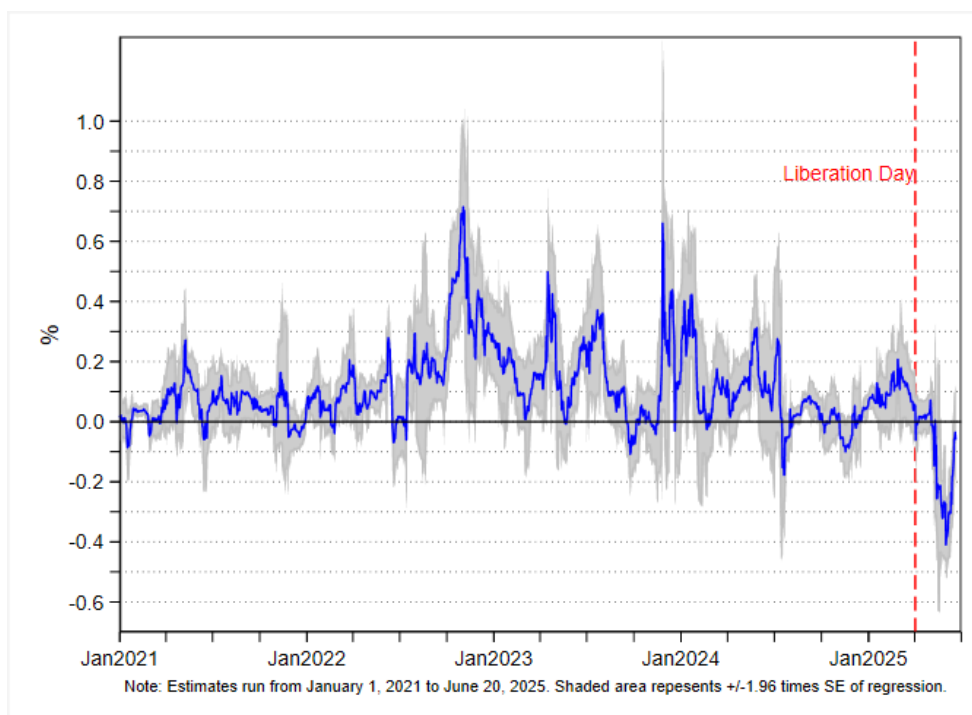


We applied the standard Chow test to assess whether a significant structural break occurred in regression #2 after Liberation Day. The Chow F-statistic was 9.76, indicating the presence of a structural break in the relationship at the 1 percent level of significance. This reinforces the message provided by the outsized residuals in Figures 3 and 4 above.

While we have confirmed a significant break in the relationship of the dollar to its primary determinants, we have not yet pinned that break on a change in the dollar's safe-haven behaviour. To address this issue, we focus specifically on the sensitivity of the dollar to changes in the VIX, as measured by the coefficient on the changes in the VIX in regression #2. We estimate this sensitivity for 30-day moving (and overlapping) windows over the course of our

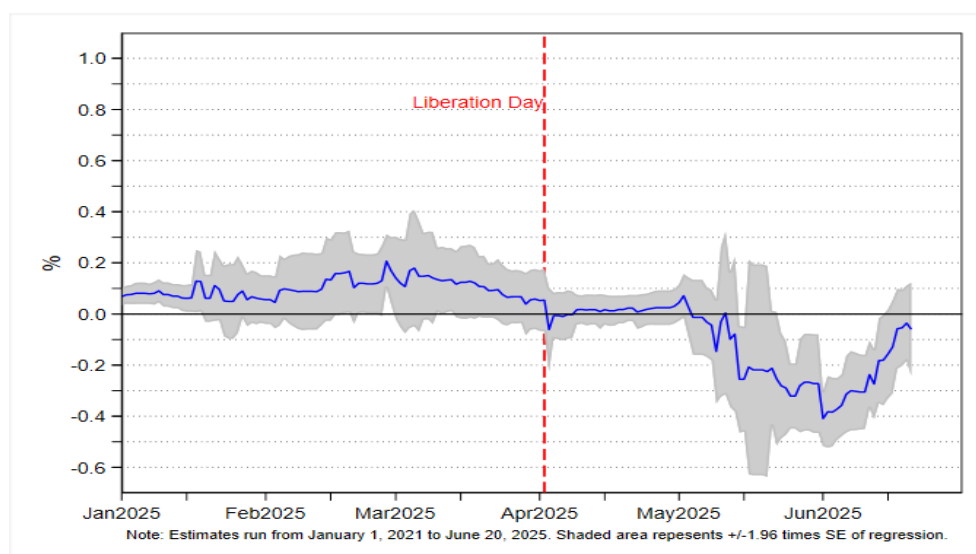
four-year estimation sample.⁴ The grey shaded areas represent 95% confidence intervals around the estimated coefficients. As indicated in the chart below, this sensitivity shifted from positive to negative in the two months after Liberation Day and reached its most negative level of the entire sample period. So, clearly, the events of Liberation Day and its aftermath exerted a profound effect on the behaviour of the dollar. However, as shown in Figure 5 and especially Figure 6, which hones in on just the past few months, the sensitivity of the dollar to the VIX started to reverse its earlier sharp decline starting in June, and as of the end-date on June 20, was insignificantly different from zero.

Fig 5. Estimated Sensitivity of the Dollar to the VIX in Regression #2



⁴ The lagged residuals from regression #1 (the levels model) are based on its estimation for the entire 2021–2025 period. The alternative—that is, estimating the levels model for 30-day periods—would lead to very imprecise estimates.

Fig 6. Estimated Sensitivity of the Dollar to the VIX in Regression #2: 2025 Only



As noted in the introduction, it is too soon to tell how the sensitivity of the dollar to the VIX will evolve going forward. It could be that as markets recover from the shock of Liberation Day, that sensitivity could return to positive territory. Or it may stay near zero, suggesting that the dollar has lost its safe-haven status but not yet become a permanent “risk-on” currency such as those of many emerging market economies. Or it may be that the safe-haven status of the dollar depends on the source of the shock to market volatility, with that status disappearing when the turbulence is caused by US policies but re-emerging when markets are jolted by geopolitical developments. Thus, the eruption of the Israel-Iran-US conflict may have pushed investors back into dollars, but that could reverse once that conflict abates.

V. Widening the Sample Period

The results described above are not an artefact of the sample period. The appendix presents the estimation results for the model when the beginning of the sample period is moved back to January 2007. As indicated in column (1) of Appendix Table 1, the

estimated coefficients from the levels regression are much larger than those shown above for the 2021-2025 sample period. As well, Appendix Figure 1 shows much larger deviations of actual and in-sample predicted values of the dollar than were evident based on the more recent sample period.

However, the coefficients on interest rates and the VIX from the daily changes regression shown in column (2) are quite similar to those from the more recent sample period. In consequence, as shown in Appendix Figure 2, the negative residuals following Liberation Day continue to be outsized, exceeded only a couple of times earlier in the sample period. A Chow test confirms, at the 1% level of significance, that a structural break in the relationship occurred after Liberation Day. By the same token, the estimated sensitivity of the dollar to the VIX, shown in Appendix Figure 3, also drops to become substantially negative, exceeded only by a drop in mid-2017, before retracing part of its decline in the past couple of weeks.

VI. Conclusion

We have confirmed anecdotal reports that in the two months since Liberation Day, the dollar switched from rising during periods of financial market volatility to falling – it thus appeared to switch from being a safe-haven currency to a “risk on” currency. That shift in investor attitudes toward the dollar was likely triggered by Trump’s chaotic tariff announcements, and perhaps reinforced by the forthcoming passage of large tax cuts that will further grow the federal deficit and debt. However, in the past couple of weeks, the sensitivity of the dollar has retraced part of its post-Liberation Day decline into negative territory. It is thus too soon to identify the events of the past few months as the beginning of the end of the dollar’s safe-haven status, or of dollar dominance more generally. But in any event, these events do caution that investors will pull back from the dollar in response to capricious,

disruptive, and counter-productive policies. And if maintained, such policies ultimately would do serious damage to the dollar's role in global trade and financial markets.

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Appendix: Estimation Results Based on 2007-2025 Sample Period

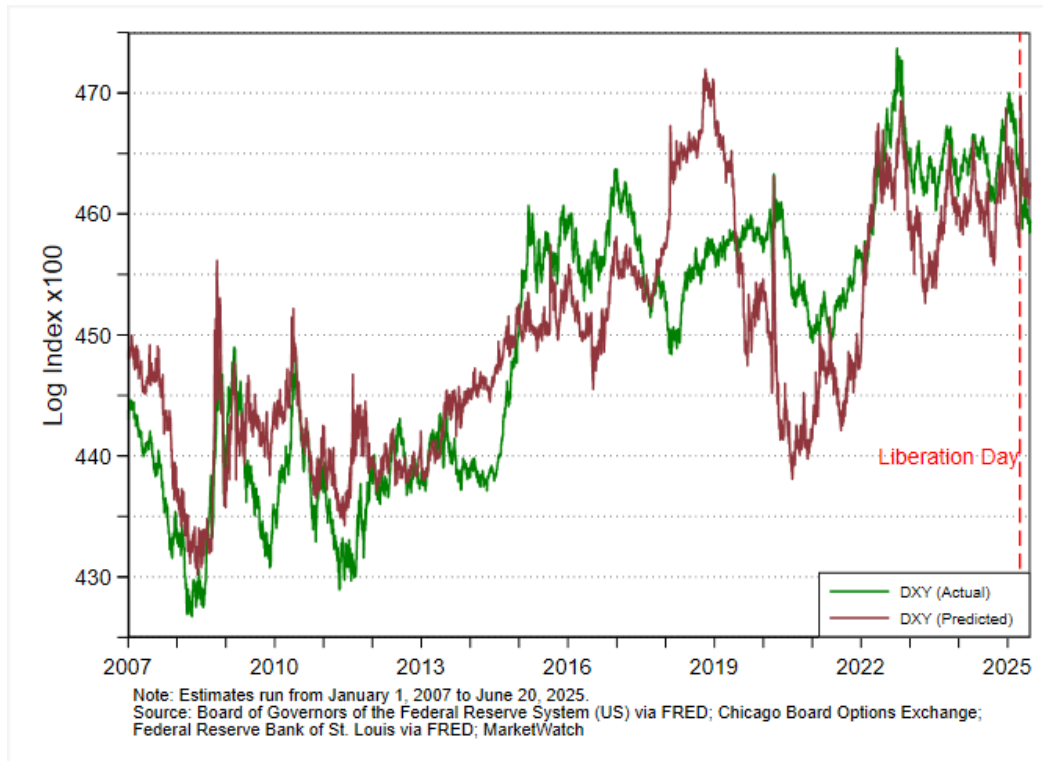
Appendix Table 1: Regressions of the DXY Dollar on its Main Determinants

VARIABLES	(1) Log DXY Index x100 OLS 2007-01-01 to 2025-06-20	(2) Log DXY Index x100 (daily changes) OLS 2007-01-01 to 2025-06-20
CBOE Volatility Index: VIX	0.31*** (0.01)	
CBOE Volatility Index: VIX, daily change		0.04*** (0.01)
2yr Yield Differential	14.12*** (0.18)	
Yield Slope Differential	9.94*** (0.27)	
2yr Yield Differential, daily change		3.23*** (0.24)
Yield Slope Differential, daily change		1.08*** (0.29)
Reg 6 Residual, 1-Day Lag		-0.00*** (0.00)
Constant	430.89*** (0.39)	0.00 (0.01)
Observations	4,400	4,391
R-squared	0.68	0.09

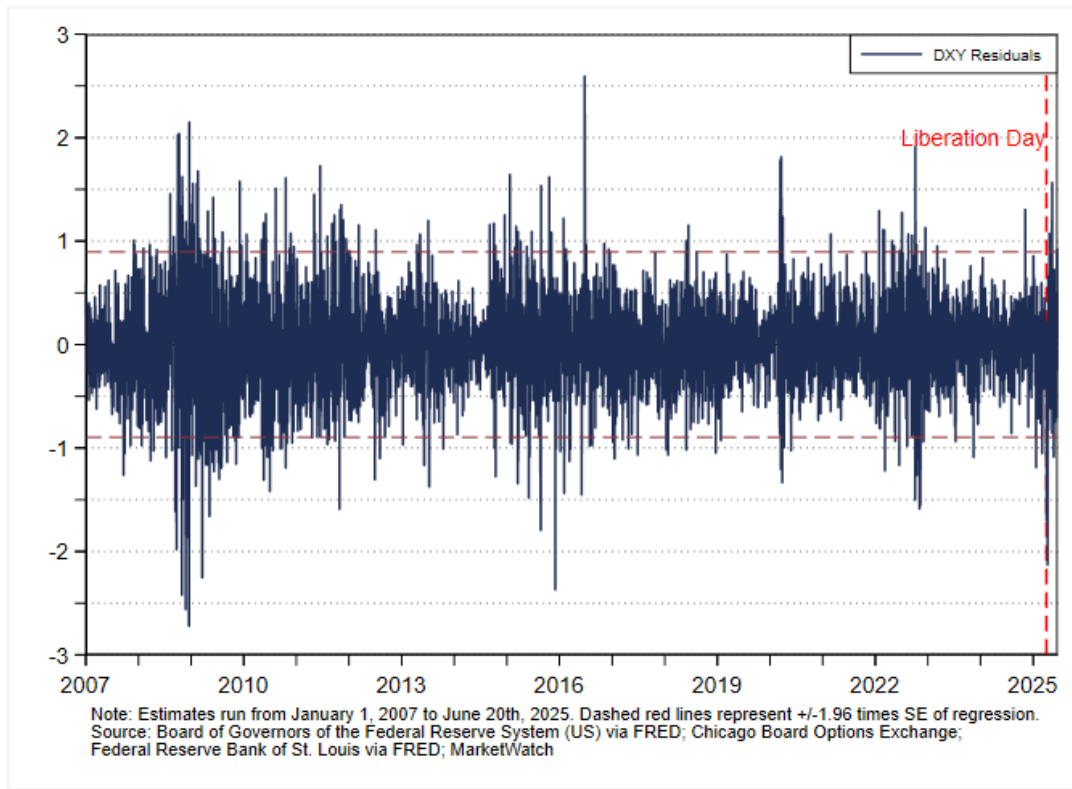
Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Appendix Figure 1: Actual and Predicted Values of the DXY Dollar Based on Levels Regression, Estimated from 2007-2025 Sample



Appendix Figure 2: Daily Changes in the DXY Index of the Dollar: Residuals Based on Regression #2, Estimated From 2007-2025 Sample



Appendix Figure 3: Estimated Sensitivity of the Dollar to the VIX in Regression #2

